COMPANY OVERVIEW

AI Data Center

What the business sells, the software it runs on, and what its data says about the facility, the plans and the fees.

Software da Database User 13

> Figures are illustrative. Jurisdictions and amounts are invented for the sample data set and are not legal, tax or payroll advice.

7	28	40	75	4	22
databases	services in 7 categories	GPUs, 36 in service	knowledge-base articles	costed business plans	government fees

1. What the business does

Software da Database User 13 is an AI data center services company. Clients rent GPU capacity to train and serve models, send inference traffic to hosted models, keep data in tiered storage, place their own hardware in its racks, and buy support from its staff. The rate card, the `service_fees` dataset, lists 28 services in seven categories:

Category	Services	What is sold	Billed per
Compute	7	Dedicated H100 80GB GPU; Dedicated A100 80GB GPU; Dedicated L40S GPU; On-demand H100 80GB; On-demand A100 80GB; On-demand L4; CPU compute node	gpu-month, gpu-hour, node-month
Inference	6	Inference, input tokens; Inference, output tokens; Embedding generation; Rerank requests; Speech-to-text; Model fine-tuning run	million-tokens, thousand-requests, audio-hour, gpu-hour
Storage	5	Block storage, hot tier; Object storage, warm tier; Archive storage, cold tier; Archive retrieval; Backup retention, off-site	tb-month, tb
Network	2	Egress bandwidth; Private cross-connect	tb, port-month
Colocation	2	Rack colocation, full cabinet; Additional power draw	rack-month, kw-month
Support	3	Remote hands; Managed support, standard; Managed support, 24/7	hour, month
Professional	3	Security audit engagement; Onboarding and migration; Compliance evidence pack	engagement, year
Total	**28**		

The company runs its own operations on the software this document describes: one program that keeps seven databases, for contacts, documents, datasets, model registry, embeddings, jobs and audit.

2. The seven databases

Each database is its own SQLite file under the storage root, so one domain can be backed up, inspected or replaced without touching the others. The order below is the order they are provisioned in. The numbers in brackets are the rows each table holds after `python -m datacenter seed` fills an empty root, which loads the bundled data and runs the jobs that loading queues. The audit count grows with every later command.

#	Database	Purpose	Tables (rows)
1	**Contacts** `contacts.db`	People and organizations the business works with.	`organizations` (20) `contacts` (45) `contact_channels` (42)
2	**Documents** `documents.db`	Documents, their text and their metadata.	`documents` (79) `document_tags` (219)
3	**Datasets** `datasets.db`	Structured data records grouped into named datasets.	`datasets` (12) `records` (462)
4	**Model registry** `models.db`	AI models available to the data center and their versions.	`models` (72) `model_versions` (79)
5	**Embeddings** `embeddings.db`	Vector index used for semantic search over documents.	`embeddings` (79)
6	**Jobs** `jobs.db`	Queue of processing work: ingest, embed, export.	`jobs` (1) `job_runs` (1)
7	**Audit** `audit.db`	Append-only log of everything that happens in the data center.	`events` (466)
	Total		**1,577 rows**

3. Architecture

The command line and the Python API both call one class, `DataCenter`, which owns the connections to all seven databases. Data files are validated in full before anything is written, then loaded through that class. Documents are embedded for search as they are stored, and every write is also recorded in the audit database. Backups copy all seven files together with a manifest.

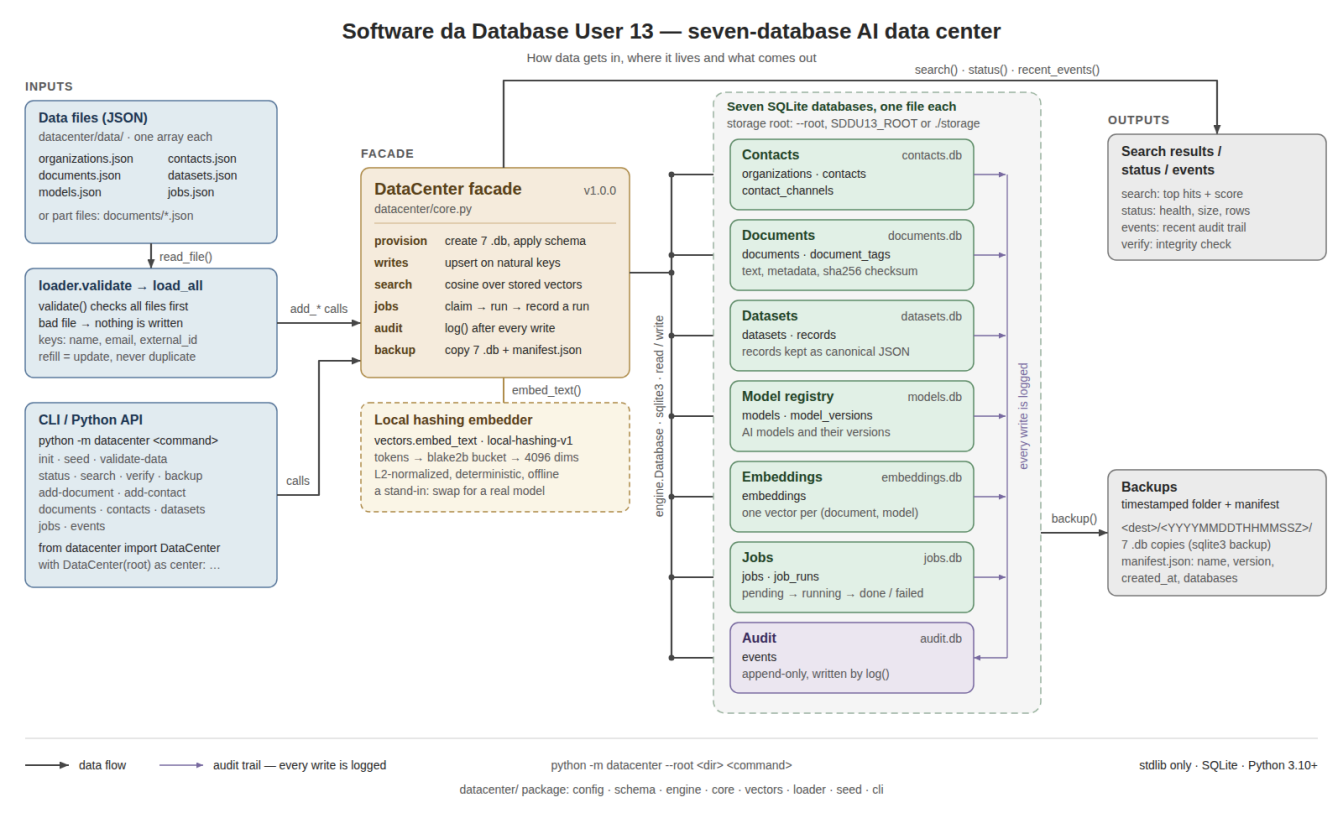


Figure 1. How data gets in, where it lives and what comes out. Inputs on the left, the `DataCenter` class in the middle, the seven database files in the dashed box, and search results, status and backups on the right. Purple arrows are the audit trail. The embedder box is out of date: an embedding is now stored as a sparse map of the hash buckets a text uses, not as a fixed-length vector.

4. Facility snapshot

GPU inventory

The `gpu_inventory` dataset lists 40 GPUs in 11 hosts across 5 racks (R01, R02, R03, R04 and STORE-A), with 2,536 GB of GPU memory in total.

Model	GPUs	Memory (GB)
H100 80GB	16	1,280
A100 80GB	8	640
L40S	5	240
A100 40GB	4	160
A10	3	72
L4	2	48
RTX 6000 Ada	2	96
Total	**40**	**2,536**

Status	GPUs
in-service	36
spare	2
repair	1
retired	1
Total	**40**

Power and cooling

The `power_and_cooling_daily` dataset holds 60 daily readings for rack R01, June 1, 2026 to July 30, 2026. Averages are daily means; PUE ranged from 1.276 to 1.409, and the peak was reached on 12 days.

17.59 kW
average draw

24.00 kW
peak draw

1.340
average PUE

22.7 / 34.5 °C
average inlet / outlet air

Incidents

The `incidents` dataset records 25 incidents opened from February 11, 2025 to August 30, 2026; 10 affected clients. The median time to resolve was 86 minutes and the longest 2,535. Sev1 is the most severe.

Severity	Incidents	Client impact	Median min.
sev1	4	4	126
sev2	9	5	142
sev3	7	1	52
sev4	5	0	32
Total	**25**	**10**	**86**

Category	Incidents	Median min.
gpu	4	68
network	4	97
software	4	34
storage	4	1,171
cooling	3	102
power	3	48
security	3	97

5. Knowledge base

The documents database holds 79 documents. Of these, 75 form the knowledge base, under four sources: what staff and clients need to know about AI, how the hall is run, the rules that govern data and models, and the terms of employment. The other 4 documents come from `internal/runbooks`, `internal/process` and `finance/2026-q1`. The embeddings database holds 79 vectors, one for every stored document, so all of them are reachable by search.

Source	Documents	Examples
`kb/fundamentals`	20	What an embedding is; Tokens and context windows; Transformers in plain language
`kb/operations`	20	GPU server intake and burn-in; Rack power budgeting; Cooling and temperature thresholds
`kb/governance`	20	Data classification; Client data handling and retention; Privacy and consent for training data
`kb/employment`	15	Employment types and contracts; Probation and confirmation; Working hours, shifts and on-call
Knowledge base	**75**	

Search with `python -m datacenter search "your question"` or the Search page of the portal.

6. Business plans

The `business_plans` dataset holds 37 lines in four plans. Each line names a fee code, and `datacenter.plans` costs it against the fee table that code belongs to:

- **Government** fees are charged each time they fall due within the plan's months: once, monthly, quarterly or annually.
- **Service** fees are the unit price times the units, with the rate card's minimum enforced. Per-month units are charged for every month of the line.
- **Employment** costs are a twelfth of base salary, grossed up by the employer tax rate, plus monthly benefits, for every person and month, plus any recruitment fee once per person.

Plan	Lines	Months	Government	Service	Employment	Total	Per month
`gpu-cluster-launch`	11	12	$19,870.00	—	$457,942.00	**$477,812.00**	$39,817.67
`managed-inference`	7	12	—	$50,524.00	—	**$50,524.00**	$4,210.33
`starter-colocation`	6	12	—	$53,388.00	—	**$53,388.00**	$4,449.00
`year-one-operating`	13	12	$4,015.00	—	$459,028.00	**$463,043.00**	$38,586.92

A dash means the plan has no lines in that fee table. “Per month” is the total divided by the plan's longest line. The plans are alternatives, not parts of one budget, so they are not added together.

7. Government fees summary

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The `government_fees` dataset lists 22 fees charged by 13 authorities in three jurisdictions: Westmarch, Northgate County and Port Lindel. Fees are charged once or on a schedule, so the tables keep the two apart. “Recurring per year” counts annual fees once, quarterly fees four times and monthly fees twelve times; “First year” adds the one-time fees to that.

By category

Category	Fees	One-time	Recurring per year	First year
Registration	3	$450.00	$295.00	$745.00
Licence	1	$0.00	$380.00	$380.00
Permit	5	$6,430.00	$0.00	$6,430.00
Inspection	1	$0.00	$310.00	$310.00
Environmental	5	$0.00	$4,555.00	$4,555.00
Utility	3	$6,800.00	$6,130.00	$12,930.00
Compliance	1	$0.00	$410.00	$410.00
Tax	3	$0.00	$1,300.00	$1,300.00
Total	**22**	**$13,680.00**	**$13,380.00**	**$27,060.00**

By jurisdiction

Jurisdiction	Fees	One-time	Recurring per year	First year
Westmarch	9	$450.00	$2,940.00	$3,390.00
Northgate County	10	$6,430.00	$4,310.00	$10,740.00
Port Lindel	3	$6,800.00	$6,130.00	$12,930.00
Total	**22**	**$13,680.00**	**$13,380.00**	**$27,060.00**

Sales and use tax registration (GOV-020) costs $0.00 and is still counted in the tables above, because it must be filed. The fee table notes: “No fee to register; filing obligations follow.”

8. How to run it

The software is one Python package, `datacenter`, with no third-party dependencies. It needs Python 3.10 or newer. Run these from the repository root:

```
python -m datacenter init      # create the databases
python -m datacenter seed      # load the bundled data files
python -m datacenter status    # health, size and row counts
python -m datacenter plans     # business plans with their totals
python -m datacenter portal    # read-only portal on http://127.0.0.1:8013
```

The databases are written to a `storage` folder in the current directory. Pass `--root DIR` before the command, or set the `SDDU13_ROOT` environment variable, to keep them somewhere else. `seed` is a refill, so running it again updates records instead of duplicating them. This document is rebuilt from the same data with `python3 tools/pdf/build.py ai_data_center`.

The portal

`python -m datacenter portal` serves the data center at `http://127.0.0.1:8013` with eight pages: Overview, Search, Documents, Contacts, Datasets, Fees, Plans and Audit. It has two fixed limits:

- **Read-only.** It answers GET requests only and has no write path. Changes go through the command line or the Python API, where the audit trail records who made them.
- **Local by default.** It binds `127.0.0.1`. It has no authentication, so it refuses any other `--host` unless `--allow-remote` is also passed.

Every command

Each line below follows `python -m datacenter.`

Usage	What it does
`init`	Create the seven databases.
`status [--json]`	Health and size of every database.
`seed [--data DATA]`	Load (or refill) the data center from its data files.
`validate-data [--data DATA]`	Check the data files without loading them.
`search [--limit LIMIT] query`	Semantic search over stored documents.
`add-document [--body BODY] [--file FILE] [--source SOURCE] [--tag TAG] title`	Store and index a document.
`add-contact [--email EMAIL] [--role ROLE] [--phone PHONE] [--organization ORGANIZATION] name`	Store a contact.
`contacts`	List stored contacts.
`documents`	List stored documents.
`datasets`	List stored datasets.
`fees [--table {employment,government,service}]`	Show the government, service and employment fee tables.
`plans`	List business plans with their totals.
`portal [--host HOST] [--port PORT] [--allow-remote]`	Serve the data center as a read-only web portal.
`plan [--json] name`	Cost one business plan line by line.
`jobs [--run]`	Show the queue, or run the next job.
`verify`	Integrity-check every database.
`backup [destination]`	Back up all seven databases.
`events [--limit LIMIT]`	Show the audit trail.